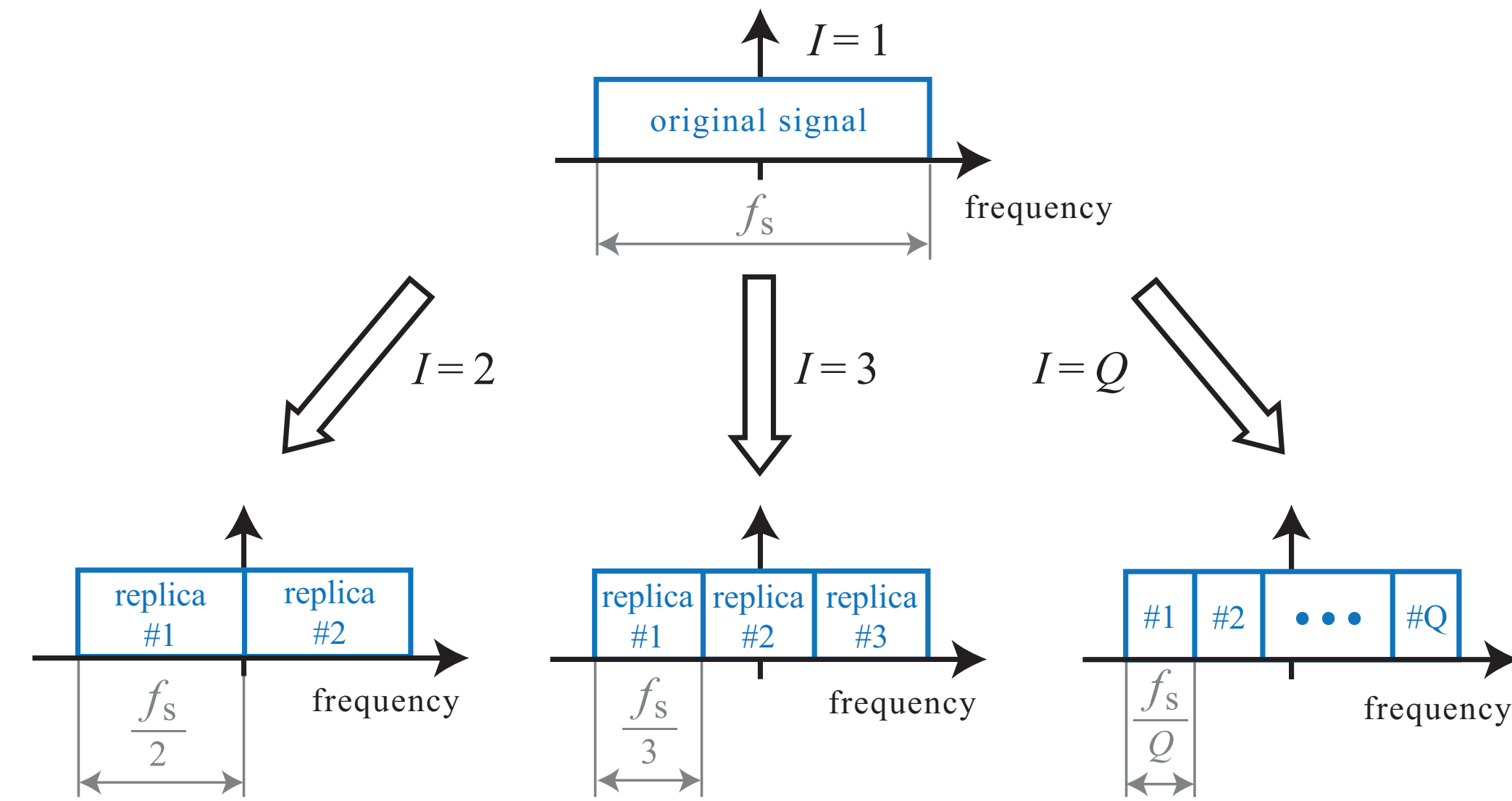


① Motivation

- high-mobility scenarios pose many challenges for performance measurements
 - expensive and complex
 - dangerous and sometimes impossible
- time-stretching (TS) technique [1]
 - conduct measurements at low speed v
 - induce effects caused by high speed $I \cdot v$
- problem:** TS technique poses unwanted side effect
 - channel estimation quality depends on the time-stretching factor I

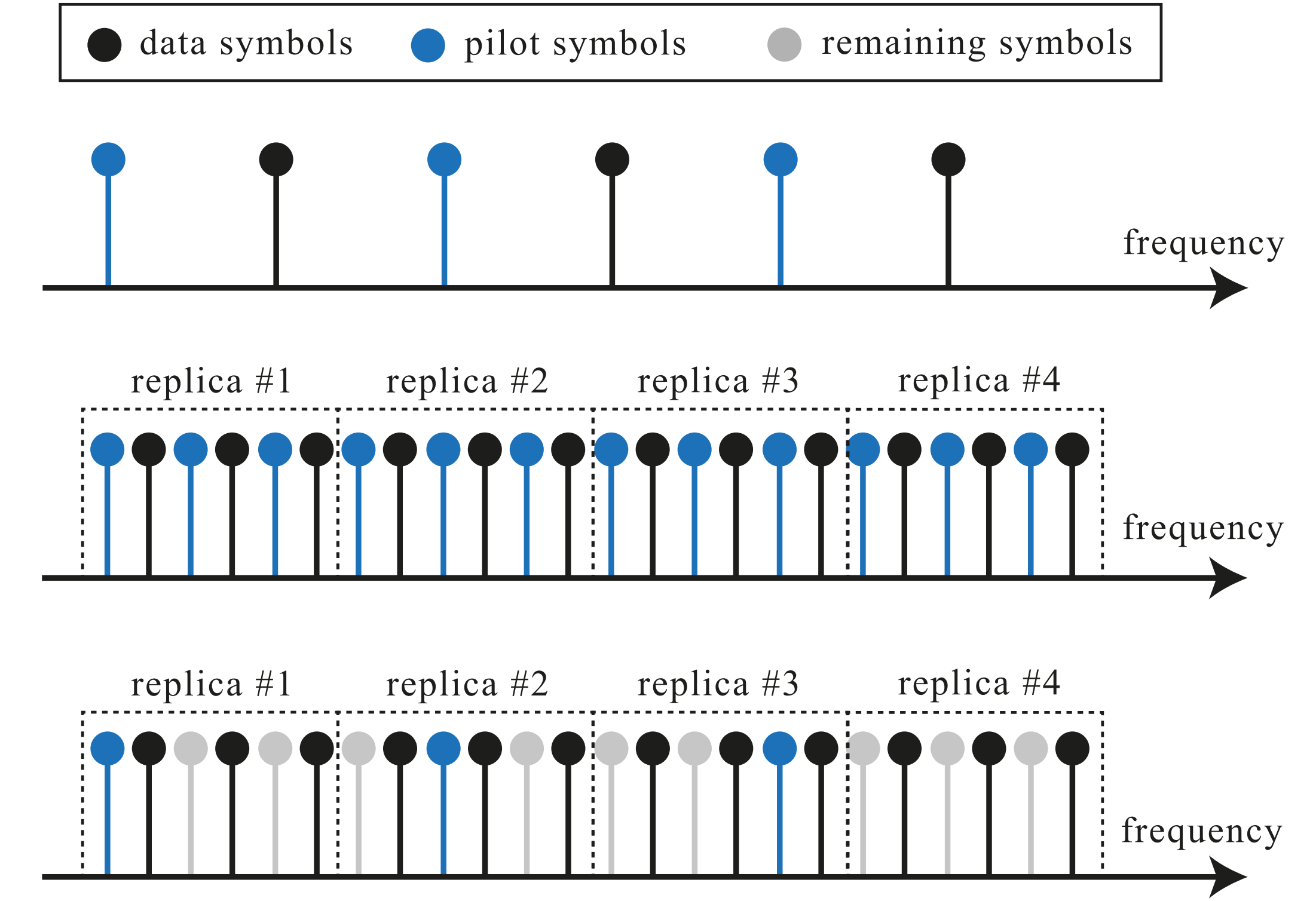
② Time-Stretching Technique

- ICI $\leftrightarrow$ normalized Doppler spread D_n
- $D_n^I = f_{D_{\max}} \cdot T^I = \frac{f_c \cdot v}{c} \cdot I \cdot T = \frac{f_c}{c} \cdot v^I \cdot T$



③ Channel Estimation

- channel usually not known at the receiver
- TS technique
 - resampling $\rightarrow$ decreased pilot spacing
 - repeating $\rightarrow$ increased number of pilots
- proposed method
 - map data symbols
 - repeat symbols in frequency domain
 - insert pilot symbols
 - consider remaining symbols as random

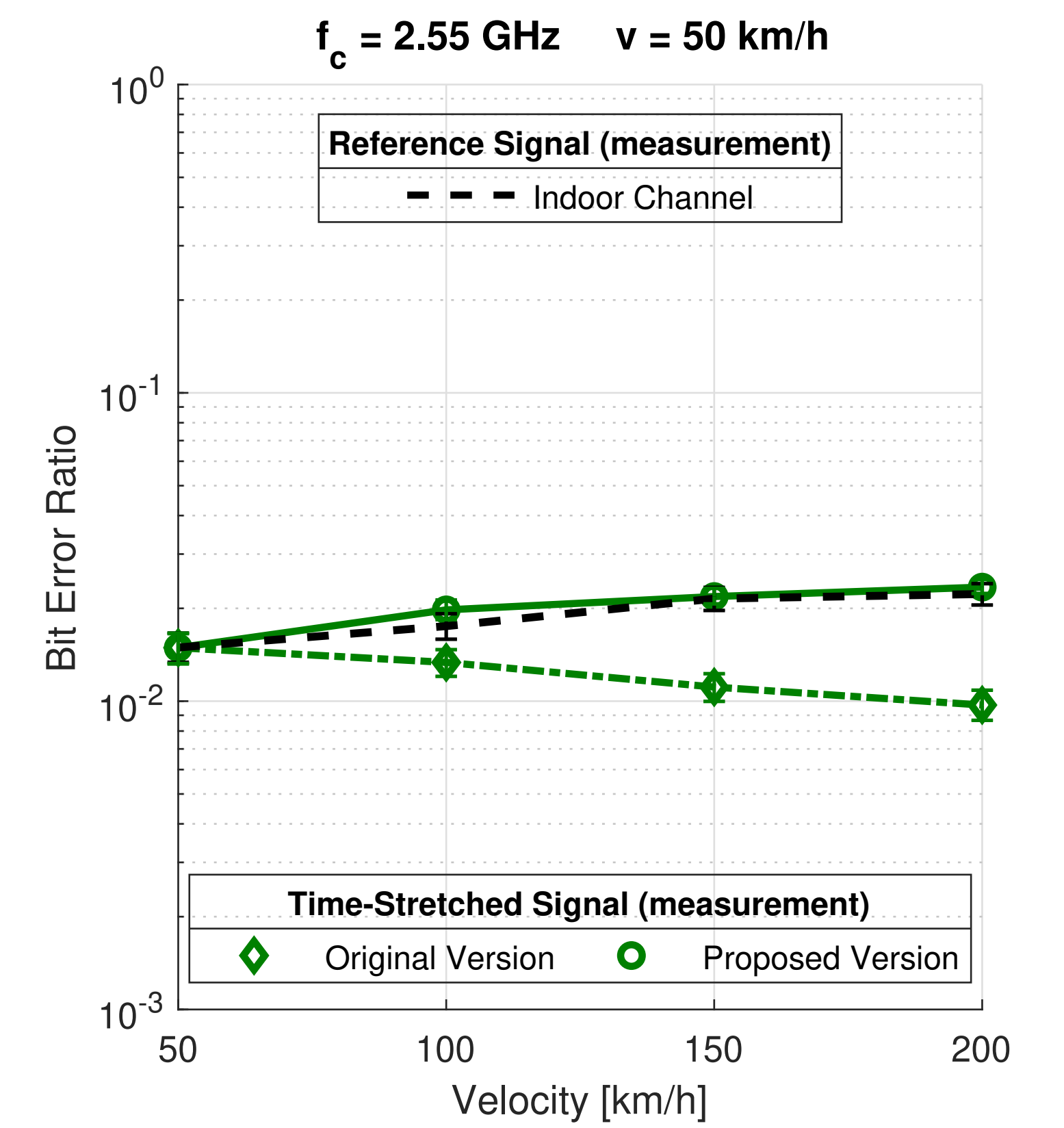
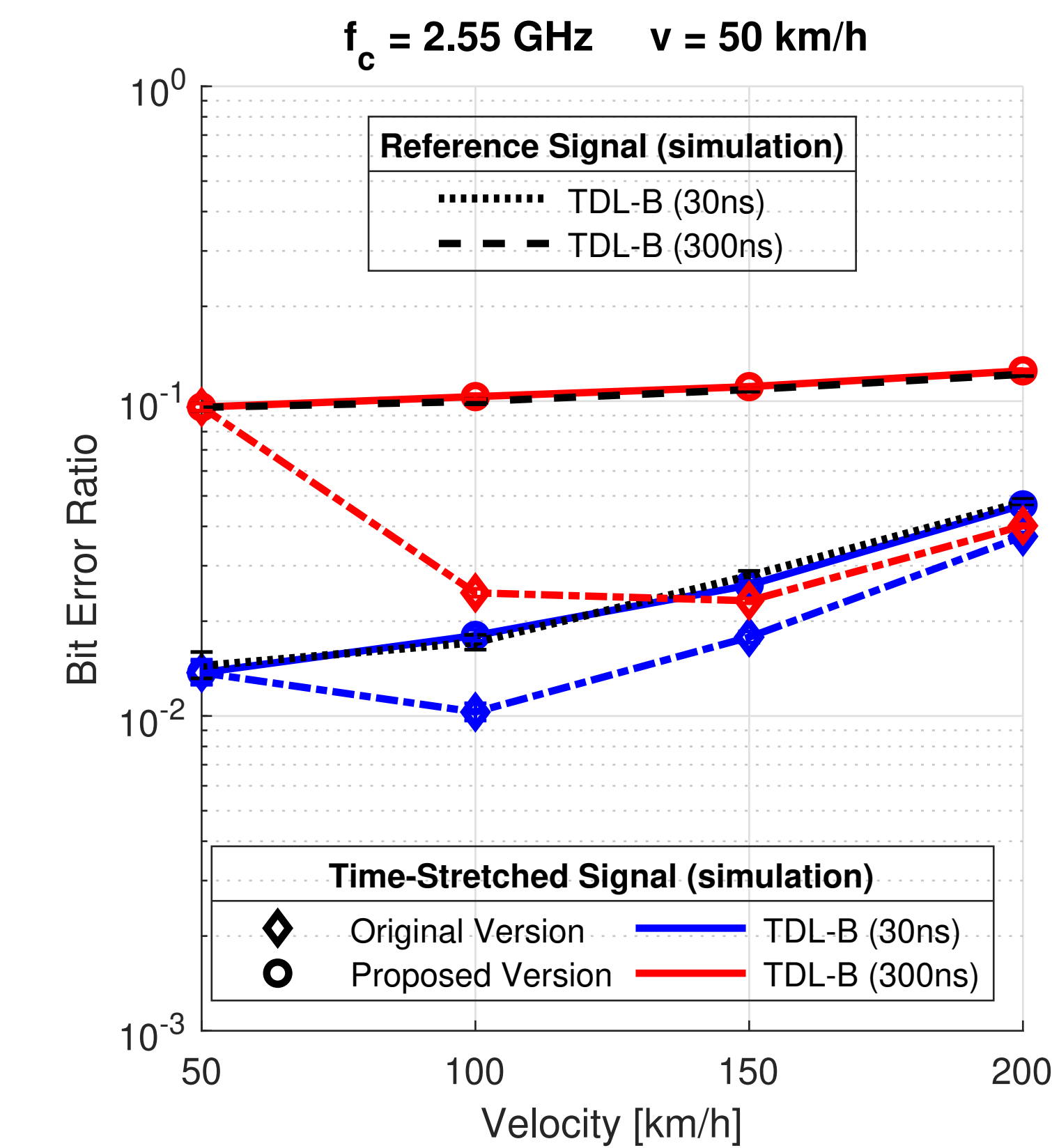
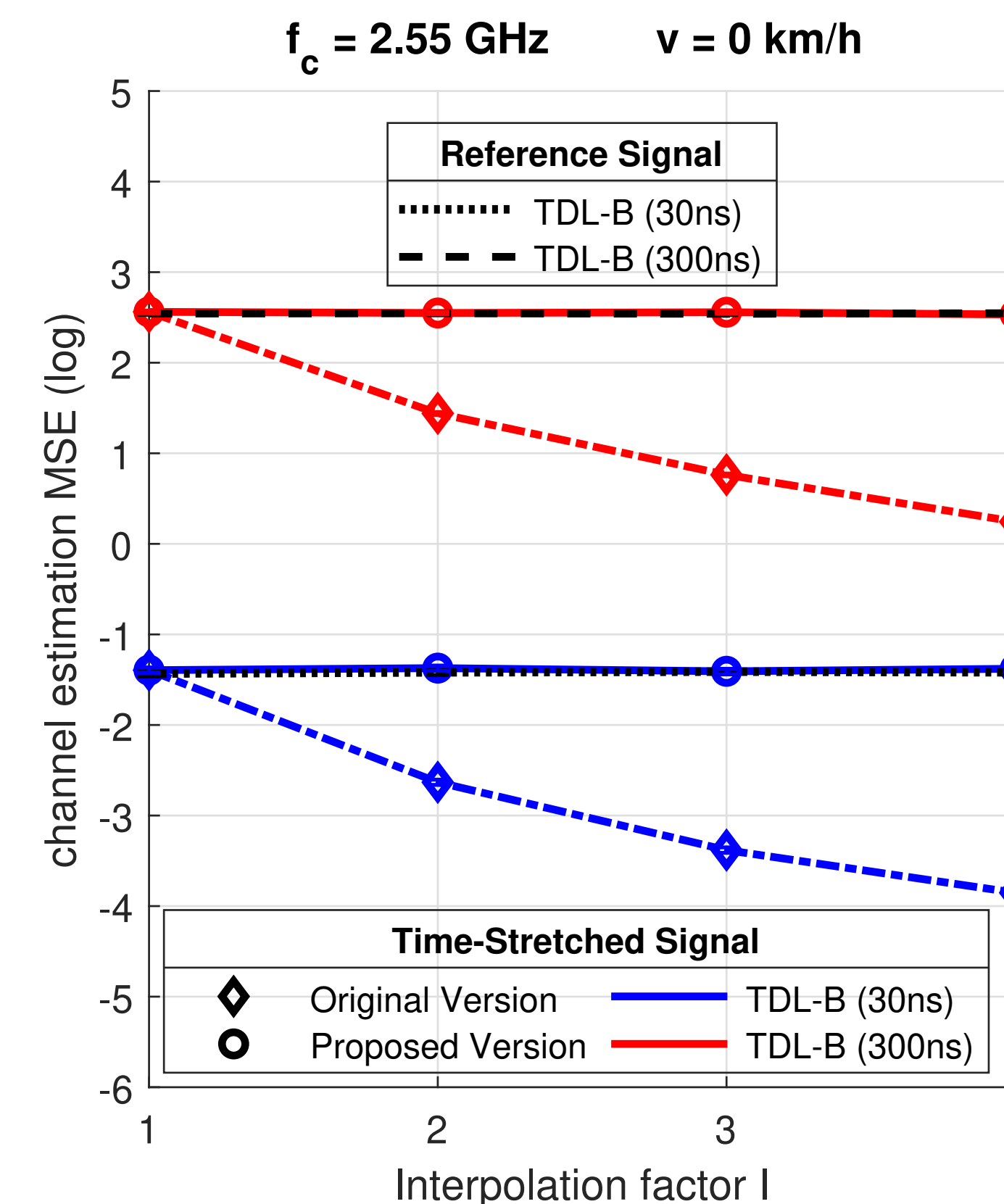


④ Measurement Procedure

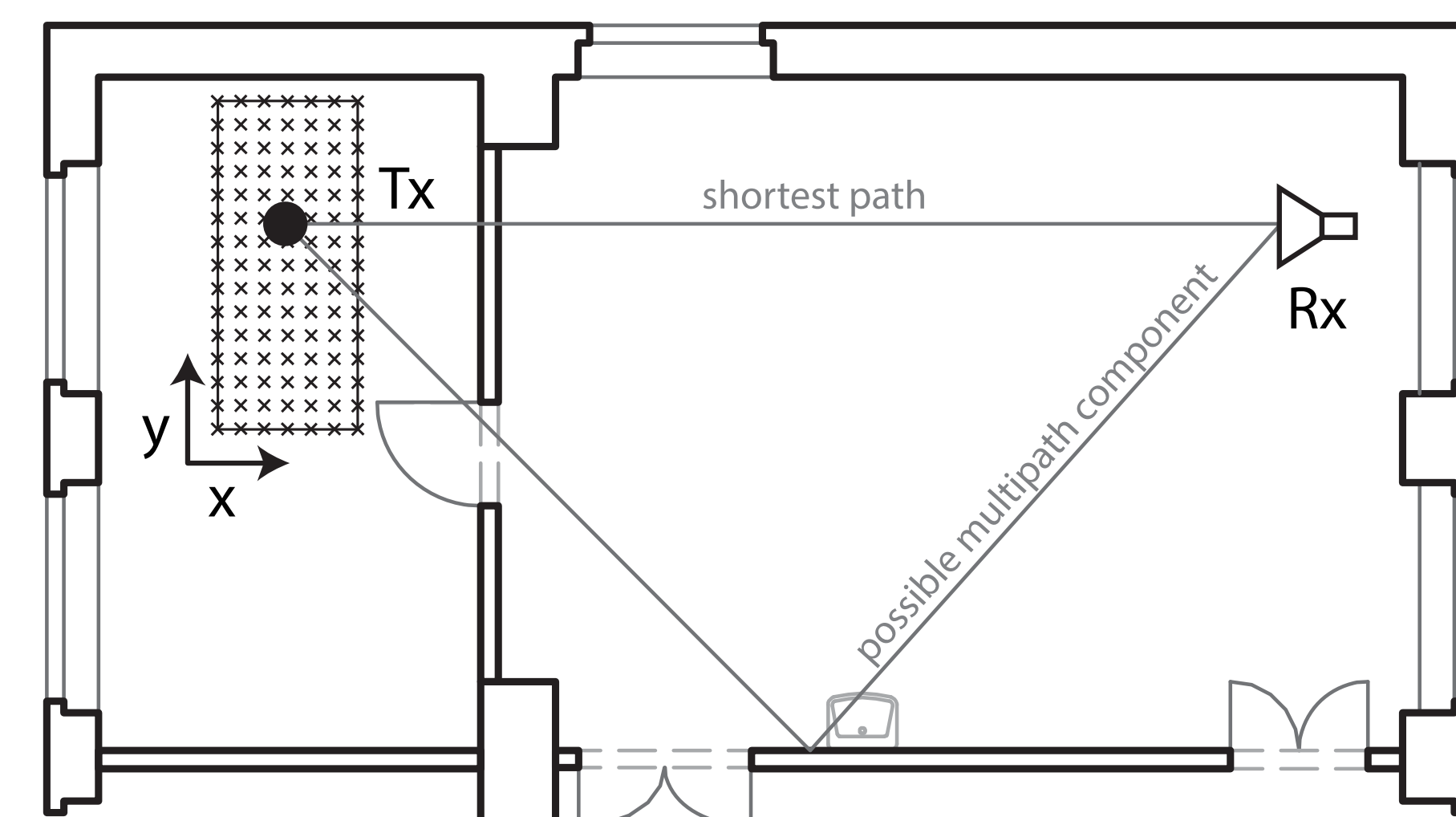
- measurement testbed described in [2]
 - rotary unit
 - repeatable and controlled measurements
- channel sounding using OFDM
 - rotating Tx and static Rx placed indoor
 - velocities of $v = 50, 100, 150$ and 200 km/h
 - 105 different positions (rectangular grid)
 - time-variant channel impulse response (CIR)

⑥ Validation

- link level simulations
 - channel estimation MSE
 - bit error ratio (BER)
 - TDL-B and measured channel
- reference signal
 - no TS technique is employed
 - behavior in the real wireless channel
- TS technique
 - actual velocity of $v = 50$ km/h
 - emulate $v = 100, 150$ and 200 km/h



⑤ Measurement Environment



⑦ Conclusion

- TS technique poses unwanted side effect
 - channel estimation quality changes over I
 - TS technique does not give accurate results
- adapted channel estimation within TS technique
 - channel estimation independent of the factor I
 - fair comparison to the original system

⑧ References

- Rodriguez-Pineiro et al., "Emulating extreme velocities of mobile LTE receivers in the downlink" EURASIP Journal on Wireless Communications and Networking (2015)
- Pasic et al., "Comparison of Sub 6 GHz and mmWave Wireless Channel Measurements at High Speeds" The 16th European Conference on Antennas and Propagation (EuCAP 2022)

